

ASTR 210 Fall 2025 — Homework set 9
50 points plus optional extra credit
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Solutions need to show all important intermediate calculations, diagrams, and explanations in addition to the final answer to receive full credit. Please check the legibility of homework solutions uploaded to the class website as we cannot grade or give credit for illegible or unreadable work. Units do not need to be specified if an expression or equation is requested; for numerical calculations please use the specified units for the answer, or if not specified, SI or cgs units. Please use Table A.1 for the value of physical constants and Table A.2 for the value of astronomical constants, unless otherwise specified in the problem. Please use the correct number of significant figures.

9.1 Extinction

The extinction in parts of the North American Nebula, in the visual band, is 1.1 magnitudes. The nebula is estimated to be at a distance of 800 pc and to be 20 pc in thickness. Suppose a main sequence B star whose absolute magnitude is $M_V = -1.1$ is observed in the direction of the nebula.

- (a) Find the apparent visual magnitude of the star if it is lying just in front of the nebula. [2 pts]
- (b) Find the apparent visual magnitude of the star if it is lying just behind the nebula. [3 pts]
- (c) If you used the apparent magnitude from (b) to calculate the distance of the star, but you forgot to correct for the nebula's extinction, what distance would you infer? What's the percentage error in the distance calculation in this case? [3 pts]

9.2 Ionizing stars and HII regions

Consider an HII region around an O6 star with a surface temperature of $T = 43650$ K and radius $R = 10.7 R_\odot$. We will integrate its spectrum to find the rate at which it produces ionizing photons and the size of its Strömgren sphere.

- (a) The integral is easier if we can work in the Wien limit ($h\nu \gg kT$). First show that will be true for this situation: find the maximum temperature that satisfies the condition $h\nu > kT$ for photons whose energy $h\nu$ is 13.6 eV and show that we can use the Wien limit for the ionizing photons from this star. $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$. [4 pts]
- (b) Find the rate Q_* at which ionizing photons are produced by this star. You'll need to do an integral, but fortunately the Wien limit allows you to use a somewhat simplified version of the Planck function. You should make use of the fact that

$$\int x^2 e^{cx} dx = e^{cx} \left(\frac{x^2}{c} - \frac{2x}{c^2} + \frac{2}{c^3} \right).$$

Hints: recall that the specific luminosity of star can be calculated as $L_\nu = (4\pi R^2)\pi I_\nu$, and see the material in pages 141-143. [7 pts]

- (c) If the density of hydrogen atoms surrounding the star is $n = 10^9 \text{ m}^{-3}$, what is the size of the star's Strömgren sphere? How does it compare to a typical distance between stars, which is on the order of 1 pc? You can assume that the recombination coefficient is $\alpha = 2.6 \times 10^{-19} \text{ m}^3 \text{ s}^{-1}$. [4 pts]

9.3 Conservation of angular momentum

- (a) Derive the result in equation 17.15 of your text by considering the collapse of a spherical object of constant density. Such an object has a moment of inertia $I = (2/5)MR^2$ and an angular momentum $L = I\omega$. You can begin your derivation by writing that the initial angular momentum equals the final angular momentum: $L_0 = L_f$. The mass of the object doesn't change, but its radius does. [4 pts]
- (b) Use equation 17.15 with 17.16 to derive 17.17 and plug numbers to verify the scaling relations. (For example: write $R_0 = (R_0/4000 \text{ AU}) \times 4000 \text{ AU}$.) [5 pts]
- (c) Enhancement: suppose the object begins as a constant density object, as described above, but suppose that it becomes more centrally concentrated as it collapses. In particular, let $I_f = (1/5)MR^2$. Re-derive a modified version of equation 17.15 for this case. [4 pts]

9.4 Post-MS nucleosynthesis

After the main sequence hydrogen-burning phase ends, helium can be burned to carbon in the triple-alpha process. To work with this process quantitatively, you'll need the following data. The mass of ^4He is 4.0026032497 u ("atomic mass unit"), where $\text{u} = 1.66053873 \times 10^{-27} \text{ kg}$. The mass of ^{12}C is 12.000000 u exactly, and that's actually the definition of u.

- (a) Calculate the energy released in this reaction and, for comparison purposes, the energy released per nucleon. (A nucleon is a neutron or a proton.) [3 pts]
- (b) If a solar-mass star burns through a core of $0.3 M_\odot$ of He at a luminosity of $25 L_\odot$, how long does the core helium burning phase last? You can assume that the core is initially 100% ^4He . [6 pts]
- (c) Further reactions beyond carbon may be available for higher mass stars. For example, ^{12}C can be burned to ^{28}Si and other isotopes in that mass range. The ^{28}Si can be burned to ^{56}Fe . If the mass of ^{28}Si is 27.976926 u and the mass of ^{56}Fe is 55.934942 u, calculate the energy released per nucleon in burning ^{12}C to ^{28}Si and then from ^{28}Si to ^{56}Fe . You should find that as you climb the periodic table towards iron, you get less and less energy per nucleon. [5 pts]

EXTRA CREDIT

9.5 Free-fall time calculation

The text uses Kepler's third law to calculate the free-fall collapse time for a molecular cloud in the process of star formation. We can also calculate the free-fall time more directly by integrating the equations of motion.

Imagine a small mass dm poised at rest, a distance R_0 from a point mass M . If we release dm from rest, how long will it take to fall into M ?

- (a) Use the principle of the conservation of energy to show that the infall speed of dm is given by the following. [5 pts]

$$v(r) = -\sqrt{2GM\left(\frac{1}{r} - \frac{1}{R_0}\right)}.$$

- (b) Write $v(r)$ as dr/dt and rearrange the previous result so that you can integrate to find $r(t)$. You should find that $r = 0$ at the free-fall time given in your text, as long as you define ρ_0 such that $M = (4\pi/3)R_0^3\rho_0$. Also note the following. [5 pts]

$$\int_0^1 \left(\frac{x}{1-x}\right)^{1/2} dx = \pi/2.$$